1. Turnaround Mechanism

- **Φ-Field Reversal:**
  - At the 25-day mark (34.25 light-years from Earth), the Φ-field generator reconfigures the spacetime curvature. The field, originally contracting spacetime ahead and expanding behind, reverses its gradient.
  - This involves switching the energy flow: the antimatter annihilation chamber redirects energy to contract spacetime toward Earth and expand it away from the destination.
  - The dark energy equivalent, providing negative pressure, adjusts to support the new curvature direction, stabilizing the bubble during the transition.
- **Time for Reversal:**
  - The Φ-field reconfiguration is instantaneous in the spacecraft's frame due to quantum field control, but external spacetime adjustment takes a brief period. Assume a 1-hour (3,600 s) transition, fitting within the 50-day total (leaving 24.5 days each way effectively).
  - The bubble's inertia (spacetime momentum) allows a smooth shift without disrupting the internal environment.
- **Energy Cost:**
  - The turnaround requires a surge of energy to reshape the field. Allocate $\frac{1.8 \times 10^{19}}{50 \times 24} \approx 1.5 \times 10^{16}$J per hour, using a fraction of the total energy (e.g., 10% of 1 hour's allocation, 1.5×10^{15}J), well within the 50-day budget.

2. Return Journey

- **Contraction for Return:**
  - The Φ-field now contracts the 34.25 light-years back to Earth into a 24.5-day equivalent distance:

$$d_{\text{contracted, return}} = \frac{34.25}{k} \approx \frac{34.25}{500} \approx 0.0685 \text{light-years},$$

    Convert to light-days:

$$d_{\text{contracted, return}} \approx 0.0685 \times 365.25 \approx 25 \text{light-days},$$

    Adjusted for 24.5 days:

$$k_{\text{adjusted}} = \frac{34.25}{24.5/365.25} \approx 510.6,$$

    This slight increase in (k) is manageable with the existing field.
- **Effective Speed:**
  - $v_{\text{eff}} = k_{\text{adjusted}} \cdot c \approx 510.6c,$
  - Travel time: $t_{\text{years}} = \frac{24.5}{365.25} \approx 0.06707 \text{years},$
  - $v_{\text{eff}} = \frac{34.25}{0.06707} \cdot c \approx 510.6c$, consistent.

- **Energy Distribution**:
  - Total energy 1.8×10^{19}J over 50 days = 4.17×10^{12}W.
  - Outbound: 25 days, 9×10^{18}J (half total, split between antimatter and dark energy).
  - Turnaround: 1 hour, 1.5×10^{15}J.
  - Return: 24.5 days, 8.85×10^{18}J (remaining energy adjusted).

3. Propulsion and Movement

- **Warp Bubble Continuity**:
  - The Φ-field maintains the bubble throughout. During turnaround, the field generator rotates the contraction-expansion gradient, guided by onboard quantum computers.
  - The spacecraft remains stationary within the bubble, and spacetime flows past it in the reverse direction.
- **Forward Motion on Return**:
  - The reversed Φ-field contracts spacetime toward Earth at $k \approx 510.6$, propelling the bubble back at (510.6c) effectively.
  - Antimatter energy sustains the field, while dark energy's negative pressure ensures stability, reducing the energy gradient needed.
- **Arrival on Earth**:
  - At 50 days, the Φ-field is deactivated, collapsing the bubble. The spacecraft re-enters normal spacetime at Earth's location, synchronized with the 50-day clock.

Detailed Mechanics

- **Field Adjustment**:
    - The Φ-field's metric $g_{\mu\nu}$ shifts from $\frac{1}{500}g_{\mu\nu}$ (outbound) to $\frac{1}{510.6}g_{\mu\nu}$ (return), recalibrated during the 1-hour turnaround.
    - Dark energy's negative mass-energy fine-tunes the curvature, minimizing energy spikes.
- **Energy Management**:
    - Antimatter is consumed at $2\mathrm{kg/day}$ (100 kg over 50 days), with dark energy providing equivalent support, ensuring a steady $4.17 \times 10^{12}\mathrm{W}$.
- **Stability**:
    - A metamaterial shell absorbs spacetime stress, with dark energy balancing tidal forces during the turnaround.

Final Solution

- **Turnaround Method**: The Φ-field reverses its contraction-expansion gradient in 1 hour, using $1.5 \times 10^{15}\mathrm{J}$, guided by quantum control.
- **Return Journey**: The field contracts 34.25 light-years to 24.5 light-days ($k \approx 510.6$), propelling the bubble at (510.6c) for 24.5 days.
- **Propulsion**: Spacetime curvature, driven by $1.8 \times 10^{19}\mathrm{J}$ from 100 kg antimatter and 100 kg dark energy, moves the bubble, returning the spacecraft to Earth at 50 days.

This method ensures a seamless round trip, leveraging the combined energy and Φ-field dynamics.